Progressive exercises

E1 · Direct application. In a space coordinate system, we consider $A(1;-1;2)$ and $B(3;0;1)$. Provide a parametric representation of the right (AB).

E2 · Application and justification. In an orthonormal coordinate system, determine a Cartesian equation of the plan through $A(1;2;3)$ and normal vector $\vec{n} = (1;0;-2)$.

E3 · Method to be organised. In a space coordinate system, the right d is given by $(x;y;z) = (t;1+t;2-t)$, $t \in \mathbb{R}$, and plan $\mathcal{P}$ by $x+2y+z=8$. Determine their intersection.

E4 · Choice of strategy. In a space coordinate system, determine the intersection of the planes of equations $x+y+z=1$ and $x-y=0$. Provide a parametric representation if the intersection is a straight line.

E5 · Synthesis or demonstration. In a space coordinate system, study the relative position of the planes d-equations 2 $x-y+z=3$ and $4x-2y+2z=5$. Justify.